

Literature Review

SLAM, Perception and ROS 2 Navigation for an Autonomous Indoor Rover

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Course: Hadoop

Paper 1

RTAB-Map as an Open-Source Lidar and Visual SLAM Library for Large-Scale and Long-Term Online Operation

1. Title of the Paper

RTAB-Map as an Open-Source Lidar and Visual SLAM Library for Large-Scale and Long-Term Online Operation

2. Authors and Publication Details

Authors: Mathieu Labbé, François Michaud

Year: 2024

Journal / Conference: arXiv preprint

Publisher: arXiv

DOI / Link: <https://arxiv.org/abs/2403.06341>

3. Introduction

Simultaneous Localization and Mapping (SLAM) is a fundamental capability for autonomous mobile robots because the robot must simultaneously estimate its own position and construct a representation of an unknown environment. The authors present RTAB-Map as an open-source SLAM framework capable of supporting different sensor configurations, including visual and LiDAR-based systems.

4. Objective

The objective is to provide a flexible SLAM framework that can operate on different robotic platforms while supporting both visual and LiDAR sensing, long-term operation, loop-closure detection, and large-scale mapping.

5. Methodology

- **Sensors/Platforms:** RGB-D cameras, stereo cameras, LiDAR sensors, and mobile robotic platforms.
- **SLAM Framework:** RTAB-Map uses appearance-based loop-closure detection combined with graph-based optimization.
- **Visual SLAM:** RGB-D and stereo visual information are used to estimate motion and construct maps.
- **LiDAR SLAM:** Laser scans can be incorporated for 2D or 3D mapping and localization.
- **Memory Management:** RTAB-Map uses memory-management techniques to support long-term operation and large environments.
- **Evaluation:** The system is evaluated quantitatively and qualitatively using several real-world datasets, including KITTI, EuRoC, TUM RGB-D, and datasets collected using a PR2 robot.

6. Results and Findings

The study demonstrates that RTAB-Map can support different SLAM configurations depending on the available sensors and computational constraints. The authors compare visual and LiDAR approaches and

show that the most suitable configuration depends on the robot, sensor setup, environment, computational resources, and required accuracy.

7. Advantages

- Supports both visual and LiDAR SLAM within one framework.
- Suitable for RGB-D cameras such as the Intel RealSense D455 used in our project.
- Supports large-scale and long-term mapping.
- Open-source and designed for integration with different robotic platforms.
- Provides loop-closure detection, which helps reduce accumulated localization drift.

8. Limitations

- Different sensor configurations have different computational requirements.
- Visual SLAM performance can degrade in environments with poor visual features.
- LiDAR-based approaches can be computationally expensive for detailed 3D mapping.
- Performance depends heavily on sensor calibration and accurate TF transformations.

9. Critical Analysis

This is one of the most directly relevant papers to our proposed autonomous rover because our system combines a LiDAR sensor with an RGB-D camera and an external IMU. RTAB-Map provides a practical framework for combining these sensing modalities rather than requiring separate SLAM systems. Its open-source implementation also makes it appropriate for deployment on embedded robotic computers.

10. Relevance to the Proposed Project

RTAB-Map is proposed as the primary SLAM and localization framework for our autonomous rover. The RealSense D455 provides RGB-D information, while the LiDAR provides additional geometric information for mapping and localization. The external IMU can provide inertial information to improve motion estimation.

11. Conclusion

This paper demonstrates RTAB-Map as a flexible open-source SLAM framework supporting visual and LiDAR sensing for autonomous robots. Its multi-sensor capabilities and compatibility with RGB-D cameras make it highly relevant to our proposed autonomous navigation system.

12. Reference (APA)

Labbé, M., & Michaud, F. (2024). RTAB-Map as an open-source lidar and visual SLAM library for large-scale and long-term online operation. arXiv preprint arXiv:2403.06341.
<https://arxiv.org/abs/2403.06341>

Paper 2

ORB-SLAM3: An Accurate Open-Source Library for Visual, Visual-Inertial, and Multimap SLAM

1. Title of the Paper

ORB-SLAM3: An Accurate Open-Source Library for Visual, Visual-Inertial, and Multimap SLAM

2. Authors and Publication Details

Authors: Carlos Campos, Richard Elvira, Juan J. Gómez Rodríguez, José M. M. Montiel, Juan D. Tardós

Year: 2021

Journal / Conference: IEEE Transactions on Robotics, Vol. 37, Issue 6

Publisher: IEEE

DOI / Link: <https://doi.org/10.1109/TRO.2021.3075644>

3. Introduction

Reliable localization is essential for autonomous robots because navigation depends on knowing the robot's position and orientation. Visual-inertial SLAM combines camera observations with inertial measurements to improve robustness and localization accuracy. ORB-SLAM3 extends previous visual SLAM systems to support visual, visual-inertial, and multi-map configurations.

4. Objective

The objective is to develop a robust SLAM system capable of operating with monocular, stereo, and RGB-D cameras, with or without inertial measurements, while maintaining accurate localization in both indoor and outdoor environments.

5. Methodology

- **Sensors:** Monocular, stereo, and RGB-D cameras with optional IMU measurements.
- **Visual Features:** ORB features are used for visual tracking and map construction.
- **Visual-Inertial SLAM:** Camera observations are tightly integrated with IMU measurements using maximum-a-posteriori estimation.
- **Multi-Map System:** Multiple maps can be maintained and later merged when previously visited areas are recognized.
- **Loop Closure:** Place recognition is used to detect previously visited locations and correct accumulated drift.
- **Evaluation:** The system is evaluated using EuRoC and TUM-VI datasets under multiple sensor configurations.

6. Results and Findings

The authors report that ORB-SLAM3 provides robust performance across different sensor configurations. The visual-inertial configuration achieves high localization accuracy, including an average accuracy of approximately 3.5 cm on the EuRoC drone dataset and 9 mm for a specific TUM-VI handheld sequence.

7. Advantages

- Supports RGB-D cameras.
- Supports external IMU integration.
- Provides visual-inertial SLAM.
- Supports multiple maps.
- Provides loop-closure and place-recognition capabilities.
- Demonstrates strong localization accuracy.

8. Limitations

- Visual SLAM depends on sufficient visual features.
- IMU calibration and synchronization are important for reliable visual-inertial operation.
- The system can require significant computational resources.
- Dynamic environments containing many moving objects can negatively affect visual feature tracking.

9. Critical Analysis

ORB-SLAM3 is particularly valuable for understanding how an external IMU can complement camera-based localization. Although our primary proposed SLAM framework is RTAB-Map, ORB-SLAM3 provides an important alternative and theoretical reference for visual-inertial localization. It also supports the decision to integrate the external IMU rather than depending entirely on the D455's internal motion sensor.

10. Relevance to the Proposed Project

The paper is relevant to the proposed rover because our system includes an RGB-D camera and an external IMU. Visual-inertial techniques can potentially improve localization when visual information alone becomes unreliable.

11. Conclusion

ORB-SLAM3 demonstrates that combining visual information with IMU measurements can provide robust and accurate localization. Its support for RGB-D cameras and external inertial sensing makes it an important reference for the localization architecture of our autonomous rover.

12. Reference (APA)

Campos, C., Elvira, R., Gómez Rodríguez, J. J., Montiel, J. M. M., & Tardós, J. D. (2021). ORB-SLAM3: An accurate open-source library for visual, visual-inertial, and multimap SLAM. *IEEE Transactions on Robotics*, 37(6), 1874–1890. <https://doi.org/10.1109/TRO.2021.3075644>

Paper 3

YOLOv7: Trainable Bag-of-Freebies Sets New State-of-the-Art for Real-Time Object Detectors

1. Title of the Paper

YOLOv7: Trainable Bag-of-Freebies Sets New State-of-the-Art for Real-Time Object Detectors

2. Authors and Publication Details

Authors: Chien-Yao Wang, Alexey Bochkovskiy, Hong-Yuan Mark Liao

Year: 2023

Journal / Conference: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

Publisher: IEEE/CVF

DOI / Link: <https://arxiv.org/abs/2207.02696>

3. Introduction

Real-time object detection is an important component of autonomous robots because the robot must identify people, vehicles, obstacles, and other objects while simultaneously performing navigation and control. The authors address the trade-off between detection accuracy and inference speed in real-time object detection.

4. Objective

The objective is to develop an object-detection architecture that achieves high detection accuracy while maintaining sufficiently high inference speed for real-time applications.

5. Methodology

- **Architecture:** YOLOv7 uses an optimized real-time object-detection architecture combined with training techniques referred to as “bag-of-freebies.”
- **Dataset:** The model is trained and evaluated primarily using the MS COCO dataset.
- **Detection:** The network predicts object locations and classes directly from images.
- **Optimization:** Architecture and training strategies are designed to improve both speed and accuracy.
- **Evaluation Metrics:** Average Precision (AP), inference speed, and computational performance are used for comparison against other real-time detectors.

6. Results and Findings

YOLOv7 achieves 56.8% AP on the MS COCO dataset for the evaluated high-performance configuration and demonstrates strong real-time performance across a wide range of inference speeds. The paper reports performance from approximately 5 FPS to over 100 FPS depending on the model and hardware configuration.

7. Advantages

- Designed specifically for real-time object detection.

- Provides a strong balance between speed and accuracy.
- Suitable as a basis for detecting people and common objects.
- Can be optimized for embedded GPU platforms.
- Provides a well-established benchmark using the MS COCO dataset.

8. Limitations

- Performance depends strongly on the GPU and model configuration.
- Small objects can be difficult to detect.
- Detection accuracy can decrease in poor lighting or visually complex environments.
- A model trained on general datasets may require fine-tuning for project-specific objects.

9. Critical Analysis

YOLOv7 provides a strong research foundation for the perception component of our rover. However, the performance reported in the paper is primarily based on more powerful GPU hardware than our embedded Jetson platform. Therefore, the actual deployed model must be selected according to Jetson inference speed and memory constraints rather than simply using the largest available model.

10. Relevance to the Proposed Project

The YOLO family will be used for real-time object and person detection from the RealSense RGB camera. The detected object can then be associated with RealSense depth information to estimate its distance and position relative to the rover.

11. Conclusion

YOLOv7 demonstrates that high-speed neural-network object detection can achieve a strong balance between accuracy and real-time performance. Its architecture provides an appropriate research reference for implementing real-time perception on the Jetson-based autonomous rover.

12. Reference (APA)

Wang, C.-Y., Bochkovskiy, A., & Liao, H.-Y. M. (2023). YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 7464–7475. <https://doi.org/10.48550/arXiv.2207.02696>

Paper 4

ROS2-Powered Autonomous Navigation for TurtleBot3: Integrating Nav2 Stack in Gazebo, RViz and Real-World Environments

1. Title of the Paper

ROS2-Powered Autonomous Navigation for TurtleBot3: Integrating Nav2 Stack in Gazebo, RViz and Real-World Environments

2. Authors and Publication Details

Authors: Athul Krishna M. J., Ajai V. Babu, Suraj Damodaran, Rekha K. James, Muhammed Murshid, Tripti S. Warriar

Year: 2024

Journal / Conference: IEEE proceedings

Publisher: IEEE

DOI / Link: <https://doi.org/10.1109/SPICES62143.2024.10779642>

3. Introduction

Autonomous navigation requires a robot to estimate its position, understand its environment, generate safe paths, and execute those paths using its drive system. ROS 2 provides a modular framework for integrating these components, while the Navigation2 (Nav2) stack provides planning, control, localization, and recovery capabilities.

4. Objective

The objective is to implement and evaluate autonomous navigation for a TurtleBot3 using ROS 2 and the Nav2 stack in both simulation and real-world environments.

5. Methodology

- **Platform:** TurtleBot3 mobile robot.
- **Middleware:** ROS 2.
- **Navigation Framework:** Navigation2 (Nav2).
- **Simulation:** Gazebo and RViz.
- **Navigation:** The robot receives target positions and uses Nav2 to generate and execute navigation plans.
- **Environment:** The system is tested in both simulated and physical environments.
- **Evaluation:** Navigation behavior and system integration are evaluated across the simulation and real-world deployments.

6. Results and Findings

The study demonstrates that ROS 2 combined with Nav2 can be used to implement autonomous navigation on a mobile robot. The integration of Gazebo, RViz, ROS 2, and Nav2 provides a consistent workflow between simulation and physical testing.

7. Advantages

- Directly uses ROS 2 and Nav2.
- Demonstrates simulation-to-real-world integration.
- Provides a modular architecture.
- Supports autonomous path planning and execution.
- RViz provides useful visualization and debugging capabilities.

8. Limitations

- The TurtleBot3 platform differs from our custom rover.
- Navigation performance depends on accurate localization and sensor data.
- The study does not directly address our particular combination of RealSense D455, LiDAR, and external IMU.
- Simulation results cannot completely represent real-world sensor noise and mechanical errors.

9. Critical Analysis

This paper is particularly relevant to our system architecture because our project also uses ROS 2 as the central middleware and Nav2 for autonomous navigation. It demonstrates that navigation can be developed and tested in simulation before deployment on the physical robot, reducing hardware-testing risk.

10. Relevance to the Proposed Project

Nav2 will form the navigation layer of our rover. SLAM/localization provides the robot pose and map, while Nav2 uses this information together with obstacle data to plan safe paths toward specified destinations.

11. Conclusion

The paper demonstrates the practical use of ROS 2 and Nav2 for autonomous mobile-robot navigation across simulated and real-world environments. It provides a strong architectural reference for implementing autonomous navigation on our Jetson-based rover.

12. Reference (APA)

Krishna, A. M. J., Babu, A. V., Damodaran, S., James, R. K., Murshid, M., & Warriar, T. S. (2024). ROS2-powered autonomous navigation for TurtleBot3: Integrating Nav2 stack in Gazebo, RViz and real-world environments. IEEE proceedings.
<https://doi.org/10.1109/SPICES62143.2024.10779642>

Paper 5

A ROS 2-Based Navigation and Simulation Stack for the Robotino

1. Title of the Paper

A ROS 2-Based Navigation and Simulation Stack for the Robotino

2. Authors and Publication Details

Authors: Saurabh Borse, Tarik Viehmann, Alexander Ferrein, Gerhard Lakemeyer

Year: 2024

Journal / Conference: arXiv preprint

Publisher: arXiv

DOI / Link: <https://arxiv.org/abs/2411.09441>

3. Introduction

Developing autonomous robots requires a reliable software architecture that can operate consistently in both simulation and physical environments. ROS 2 provides a modular communication framework, while simulation environments allow navigation algorithms to be developed without risking physical hardware.

4. Objective

The objective is to develop a ROS 2 integration and simulation environment for the Robotino platform and provide a navigation configuration based on existing ROS 2 packages, particularly the Nav2 stack.

5. Methodology

- **Platform:** Festo Robotino mobile robot.
- **Middleware:** ROS 2.
- **Simulation:** Webots simulation environment.
- **Sensors:** LiDAR sensors are incorporated into the simulated platform.
- **Navigation:** ROS 2 Navigation2 packages are used for localization and autonomous navigation.
- **Hardware Integration:** The system is tested on physical Robotino platforms.
- **Evaluation:** Simulation results are compared against real-world experiments conducted in a logistics environment.

6. Results and Findings

The study demonstrates the feasibility of using ROS 2 and Nav2 for autonomous navigation and reports consistency between simulation and real-world experiments. The work shows that a simulation environment can be used as a practical development and testing platform before deploying the navigation system on physical hardware.

7. Advantages

- Combines ROS 2, simulation, LiDAR, and Nav2.
- Provides a simulation-to-real-world workflow.
- Useful for testing navigation without risking the physical robot.

- Demonstrates real-world validation.
- Uses existing ROS 2 packages rather than requiring an entirely custom navigation framework.

8. Limitations

- The Robotino platform differs mechanically from our rover.
- The study focuses primarily on navigation rather than neural-network perception.
- Webots is different from the simulation environment planned for our project.
- Real-world performance still depends on sensor calibration and robot-specific parameters.

9. Critical Analysis

This paper is valuable because it supports the development strategy of separating simulation and physical implementation. Our project uses an embedded Jetson computer and multiple sensors, so testing ROS 2, mapping, localization, and navigation in simulation can reduce the amount of trial-and-error required on the physical rover.

10. Relevance to the Proposed Project

The paper supports our proposed workflow of developing the rover's ROS 2 architecture, robot model, localization, SLAM, and Nav2 navigation in simulation before transferring the system to the physical platform. It is particularly relevant to our use of LiDAR-based navigation and ROS 2.

11. Conclusion

The paper demonstrates that a ROS 2 and Nav2-based navigation stack can be developed and validated through simulation and subsequently deployed on a physical mobile robot. This provides a practical development methodology for our autonomous rover project.

12. Reference (APA)

Borse, S., Viehmann, T., Ferrein, A., & Lakemeyer, G. (2024). A ROS 2-based navigation and simulation stack for the Robotino. arXiv preprint arXiv:2411.09441. <https://arxiv.org/abs/2411.09441>